🧠 XAI: Chess Knowledge in AlphaZero & Interpreting Stable Diffusion

FAMAF - UNC

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eXplainable
Artificial Intelligence

🧉 Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

https://interpretable-ml-class.github.io/

Acquisition of Chess Knowledge in AlphaZero

McGrath, Kapishnikov, Tomašev, Pearce, Hassabis, Kim, Paquet & Kramnik (2022)

🧉 Learning from Machines: Three-Pronged Acquisition ---

- **Probe for concepts**
  - How closely is AlphaZero's internal representation **related to** chess concepts humans have already created?
  - Detection of human concepts from network activations
- **Study behavioral changes**
  - How do changing representations give rise to changing behaviors?
  - Evolution of AlphaZero vs. human's strategy in openings
- **Investigate activations directly**
  - Unsupervised methods using non-negative matrix factorization (NMF) and direct measure of covariance to discover concepts not represented by humans in the past

🧉 Interpretability ---

- Concept-based (post-hoc) interpretability
  - Network probing / important features / mechanistic understanding
  - Challenges: correlation not causal
- Explainability in reinforcement learning
  - Structural causal models / reward difference explanations
  - Identify interesting points in behavioral trajectories

🧉 Chess as Testing Ground for AI Interpretability ---

Can humans learn the machine's strategy?

- Does not rely on human-labeled data
- Tree-based organization / saliency maps
- Natural language processing to generate move-by-move commentary
- **This paper:** captures "intuitive" aspect of chess play by understanding networks that produce value assessment (v) and candidate move ($\mathbf{p}$)

$$\mathbf{p},\ v = f_\theta(\mathbf{z}^0)$$

🧉 AlphaZero: Network Structure and Training ---

$$\mathbf{p},\; v = f_\theta(\mathbf{z}^0)$$

$$\mathbf{z}^l = f^l(\mathbf{z}^{l-1}) = \mathrm{ReLU}\Big(\mathbf{z}^{l-1} + g^l(\mathbf{z}^{l-1})\Big)$$

$$\mathbf{z}^l = f^{1:l}(\mathbf{z}^0) = f^l \circ \cdots \circ f^2 \circ f^1(\mathbf{z}^0)$$

Probing for Concepts

🧉 Encoding of Human Conceptual Knowledge ---

Question: Can human concepts be easily predicted from the network's internal representation?

Concept: User-defined function mapping network input to real line:

$$c(\mathbf{z}^0) = \begin{cases} 1 & \text{if } \mathbf{z}^0 \text{ contains a bishop-pair for the playing side} \\ 0 & \text{otherwise} \end{cases}$$

Approach: Train a sparse linear regression model from activations $\mathbf{z}^l$ at layer l and training step t to human concept j

🧉 Probing Concept Learning: Details ---

- **Data:** Randomly sample training, validation, test data from ChessBase archive, compute concept values and AlphaZero activations
- **Procedure:** For concept i, layer l, training step t, solve:

$$\mathbf{w}_{jlt},\ b_{jlt} = \min_{\mathbf{w},b} \frac{1}{N} \left\| \mathbf{w}^T \mathbf{Z}_t^l + b\mathbf{1} - \mathbf{c}_j \right\|_2^2 + \lambda \|\mathbf{w}\|_1 + \lambda |b|$$

- **Controls:** Regression from $\mathbf{z}^0$ and random concept regression
- **Evaluation:** R^2 value, fraction of variance in concept explained by network activation

🧉 Evolution of Human Concepts in AlphaZero ---

Key Findings ---

1. **Grokking:** Many concepts begin to increase in accuracy around 32,000 steps
2. Drop in linearly-available information in later layers for some concepts
3. Some concepts **cannot** be regressed: sparsity partially obstructs ability to relate **highly-distributed representations** to concepts
4. **Learning from prediction errors:** regression errors may point to a "difference of opinion" with Stockfish

🧉 Challenges for Concept Probing ---

1. What's the right **probing architecture**?
2. How should we interpret **complex or subjective concepts**?
3. When can we definitively say a **concept is represented**?
4. When we train a probe, we cannot tell if we are getting a **confounder or the concept** itself

Progression through AlphaZero and Human History

🧉 Progression of Knowledge ---

- Recall: $\mathbf{p},\ v = f_\theta(\mathbf{z}^0)$
- **Question:** How does the progression of AlphaZero's knowledge compare to that of humans?
- **Key Findings:**
  - AlphaZero: Start from a uniform prior, then narrow down.
  - Humans: Start from a concentrated prior, then expand.

🧉 Progression through Human History ---

🧉 Progression through AlphaZero History ---

🧉 Progression of AlphaZero's Chess Knowledge ---

- **Methodology:** At different training steps (up to 128k), examine AlphaZero's move tendencies and concept encodings

Primary Takeaways

1. AlphaZero learns standard opening theory early on
2. AlphaZero learns material values before more complex positional concepts

Both reinforce idea that AlphaZero learns basic human chess concepts first

🧉 Opening Theory Knowledge ---

1. ∼30–60k: AlphaZero plays *1. e4/d4* the majority of the time
2. ∼45k: AlphaZero considers *2. d4* before opting for *2. Nf3*
3. ∼45k: AlphaZero plays standard responses to *1. e4*

Material vs. Positional Knowledge ---

1. ∼30k training steps: Piece Values develop, converge ∼100k
2. King Safety, Mobility concepts emerge after Material
   - More complex concepts require more training time

🧉 Training Progression Assessment: GM Vladimir Kramnik ---

1. **16k to 32k:** Material Value in Complex Positions
2. **32k to 64k:** King Safety in Imbalanced Positions
3. **64k to 128k:** King Safety & Material Sacrifices in Complex Positions

Tactical skills appear to **precede positional skills** as AlphaZero learns

Exploring Additional Feature Detectors in AlphaZero Network

🧉 Exploring Activations with Unsupervised Methods ---

Goal: Find Feature Detectors embedded within Network

Methods:

a) Non-Negative Matrix Factorization of each layer's channels

b) Correlation of Input Board with each channel's activations

Primary Takeaway

Individual network layers & channels encode *feature detectors* related to human-recognizable chess concepts

🧉 Approach #1: NN Matrix Factorization Analysis ---

For each layer l with C channels: 1. Compute a matrix factorization $\cdot \times \mathbf{F}$ using $K < C$ columns

For each factor k $(1 \ldots K)$ and input n $(1 \ldots N)$: 2. Visualize activations on Chess Board to find *feature detectors*

$$\hat{\mathbf{Z}}^{l} \in \mathbb{R}^{NHW \times C} \qquad \cdot_{\text{all}} \in \mathbb{R}^{NHW \times K} \qquad \mathbf{F} \in \mathbb{R}^{K \times C}$$

$$\mathbf{F}^{*},\ \cdot^{*}_{\text{all}} = \min_{\mathbf{F},\ \cdot_{\text{all}}} \left\| \hat{\mathbf{Z}}^{l} - \cdot_{\text{all}} \mathbf{F} \right\|_2^2 \qquad \mathbf{F},\ \cdot_{\text{all}} \geq 0$$

🧉 Results: NN Matrix Factorization Analysis ---

🧉 Approach #2: Input-Activation Covariance Analysis ---

For each layer l and channel i: 1. Compute the covariance between input $\mathbf{z}^0$ and position activations 2. Visualize covariances on Chess Board to find *feature detectors*

$$\mathrm{cov}(z_i^l, \mathbf{z}^0) = \mathbb{E}\left[z_i^l \, \mathbf{z}^0\right] - \mathbb{E}\left[z_i^l\right] \mathbb{E}\left[\mathbf{z}^0\right]$$

🧉 Results: Input-Activation Covariance Analysis ---

Detecting Move-Types from a Square:

- 1–2) Diagonal-Attacking Pieces (Queen, Bishop)
- 3–4) Horizontally-Attacking Pieces (Queen, Rook)
- 5) Both

🧉 Conclusion: Key Findings (Paper 1) ---

1. Human-Defined Concepts can be regressed from the AlphaZero network
   - Despite never being trained on a human game of chess
2. As training progresses, AlphaZero understands basic concepts (openings, material) before more complex ones (king safety, mobility)
3. Feature Detectors of human-recognizable chess concepts are encoded by individual layers + channels in AZ Network
   - Can be found using supervised & unsupervised techniques

🧉 Limitations ---

1. Challenges for Concept Probing (previous section)
2. Knowledge acquisition is not complete — only a very small part of the model
3. Interpreting "Feature Detectors' ' found via Unsupervised techniques
   - Inherently subjective
4. No Causal Insight into the Learned Concepts (only correlation)

🧉 Future Research Areas ---

1. Addressing Concept Probing Limitations (more than sparse LR)
2. Can we go beyond finding human knowledge embedded in AlphaZero and understand what new concepts are learned?
   - Further analysis of feature detectors found with unsupervised techniques
3. How do we generalize these findings to other machine learning settings? Could we find human-recognizable concepts in different trained models?

🧉 Discussion (Paper 1) ---

- How is this paper different from methods we discussed so far?
  - In terms of goals / techniques / specific models to explain
- Does this approach have potential applications for a more general setting other than board games?
  - What are the pros/cons of using games/artificial environments as baseline for evaluating RL algorithms?
- Can this be applied to our practice of doing science?
  - Can we recreate or extract theories using similar frameworks (AI for science)?
- How should we define or understand “knowledge’ ’ in these settings?
- How convinced are you about the “interpretability’ ’ aspect?
  - Who would be potential audience that could benefit from such analysis?

What the DAAM: Interpreting Stable Diffusion Using Cross Attention

Tang, Liu, Pandey, Jiang, Yang, Kumar, Stenetorp, Lin & Ture (2023)

Presented by: Alex Lin, Jason Jabbour, Mark Mazumder

Outline ---

- Related work
- Background: Stable Diffusion
- **DAAM** for text-to-image **attribution**
- Results
  - Attribution quality analysis
  - Syntax to pixels
  - Entanglement
- Limitations & discussion

DAAM estimates per-pixel attribution for each word in a prompt (post-hoc)

🧉 Related Work ---

This work applies existing techniques (cross-attention) to an **open source, SOTA** diffusion model to probe limitations

- **Textual perturbation** (Wallace et al., 2019), Attentional Visualization, information bottlenecks to relate important input tokens to outputs of large LMs
- Probing **vision transformers for verb understanding** (Hendricks & Nematzadeh, 2021)
- Enhancing diffusion models using **prompt engineering** (Hertz et al., 2020; Woolf, 2020)
- **Disentangling** e.g., style and spelling (Karras et al., 2019; Materzynska et al., 2022)
- Counterfactual explanations for *text classification* (Jacovi et al., 2021) — unclear how to extend to image generation

🧉 Background: Generative Models ---

🧉 Background: Image Generation with Diffusion ---

🧉 Background: UNet Architecture ---

Background: Forward and Reverse Process ---

🧉 Diffusion Model Loss ---

$$L_{DM} = \mathbb{E}_{x,\, \epsilon \sim \mathcal{N}(0,1),\, t}\left[\| \epsilon - \epsilon_\theta(x_t,\, t) \|_2^2 \right]$$

🧉 Latent Diffusion Model Loss ---

$$L_{DM} = \mathbb{E}_{x,\, \epsilon \sim \mathcal{N}(0,1),\, t}\left[\| \epsilon - \epsilon_\theta(x_t,\ t) \|_2^2 \right]$$

$$L_{LDM} := \mathbb{E}_{\mathcal{E}(x),\, \epsilon \sim \mathcal{N}(0,1),\, t}\left[\| \epsilon - \epsilon_\theta(z_t,\ t) \|_2^2 \right]$$

- $\mathcal{E}(x)$: latent space encoder
- z_t: latent representation of image at time t
- Easier to estimate noise in latent space!

(Rombach et al. 2022)

🧉 Stable Diffusion Architecture ---

🧉 Cross-Attention for Text Conditioning ---

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) \cdot V$$

$$Q = W_Q^{(i)} \cdot \varphi_i(z_t), \quad K = W_K^{(i)} \cdot \tau_\theta(y), \quad V = W_V^{(i)} \cdot \tau_\theta(y)$$

- **Queries:** image tokens (UNet block i)
- **Keys, Values:** prompt tokens (text encoder τ_θ)

DAAM estimates **per-word attribution** via this cross-attention **for each subset** of prompt tokens

🧉 DAAM: Per-Word Heatmaps ---

🧉 DAAM: Attention Maps Formula ---

🧉 Results: Attribution Analysis Part 1 -- Object Attribution ---

We can evaluate DAAM as an image-segmentation tool

🧉 DAAM Segments Stable Diffusion Images ---

- DAAM does not require explicit segmentation labels
- DAAM is "open vocabulary" — can segment for **any text input** (not limited to known classes)
- Competitive with unsupervised methods; lower than supervised

🧉 Results: Attribution Analysis Part 2 -- Generalized Attribution ---

DAAM can segment **beyond nouns**:

🧉 Evaluating DAAM with a User Study ---

🧉 Visuosyntactic Analysis ---

Key Question

How does **syntax** relate to the generated **pixels**?

- Study **head-dependent pairs** in sentences using Universal Dependencies (UD) syntax
- For each dependency relation, measure how much the attribution map of the *dependent* overlaps with that of the *head*
- 3 measures: mIoU, mIoD (Mean IoD over the **D**ependent), mIoH (Mean IoD over the **H**ead)

🧉 Visuosyntactic Analysis: Head-Dependent Pairs ---

🧉 Visuosyntactic Analysis: Measures of Overlap ---

Measures of Overlap

- Mean Intersection over Union (mIoU)
- Mean Intersection over the Dependent (mIoD)
- Mean Intersection over the Head (mIoH)

mIoD $>$ mIoH: dependent's map **subsumes** head's

mIoD $<$ mIoH: head's map **subsumes** dependent's

🧉 Visuosyntactic Analysis: Results ---

🧉 Visuosyntactic Analysis: Key Takeaways ---

- **Noun Compounds** ("ice cream''): No dominance — complement one another
- **Punctuation & Articles** ("the donut''): No dominance — little semantic meaning
- **Verb & Subject** (*bird stands*): Head verb **dominates** — contextualises subject/object in surroundings (*semi-intuitive*)
- **Nominal Dependents** ("pile of oranges''): Head dominates — *intuitive*
- **Adjective Modifiers** ("wooden bench''): Dependent **dominates** — *counter-intuitive*

Summary

Attribution map of the **dependent subsumes** that of the head, and vice versa for others. Dominance is intuitive in some cases but **counter-intuitive** in others.

Visuosemantic Analysis: Cohyponym Entanglement ---

Key Question

Do semantically **similar words** have **worse** generation quality?

Cohyponym: “a giraffe and a zebra”

- SD image generation **worsens**
- Generates **one** noun but **not both**
- Attribution maps **overlap** → Feature Entanglement

Non-Cohyponym: “a zebra and a fridge”

- Generates **both** nouns
- Attribution maps are **distinct**

🧉 Visuosemantic Analysis: Adjectival Entanglement ---

Setup

Prompt: `"<adj> <noun> <verb phrase>"` (e.g., “a [rusty] shovel sitting in a clean shed”)
If **no entanglement**, background should **not gain attributes** pertaining to that adjective

Takeaway

Attribution maps for adjectives attend **too broadly** across images, beyond the nouns they modify $\rightarrow$ **Feature Entanglement**

🧉 Summary ---

- DAAM provides pixel-level attribution maps for Stable Diffusion, a state-of-the-art text-to-image generator
- These maps appear to be informative, as evaluated through segmentation tasks and user study
- DAAM can be a useful tool for further understanding and analyzing Stable Diffusion — e.g. through visuosyntactic analysis and visuosemantic analysis

🧉 Class Discussion ---

- Does DAAM give a clear understanding about how a large-scale latent diffusion model synthesizes text to image and which parts of an image are influenced the most?
- Does DAAM explain all the dynamics of how images are synthesized? If not, how should DAAM be modified to better explain image generation?
  - Other explanatory tools besides **attention** and **segmentation proposals**?
- DAAM pointed out failure cases of Stable Diffusion. Are there further interpretability methods needed to understand why **feature entanglement** is occurring and how it could be improved?

Thank You!

🧉 References --- |

Rombach, Robin, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. "High-Resolution Image Synthesis with Latent Diffusion Models." In *Proceedings of Cvpr*.